Report R10 — What Is the KEAP1 Methylation Phenotype?

Characterising R08's positive finding, and resolving the confound R08 declared unresolvable.

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Summary

R08 reported that 29.4% of probes are differentially methylated by KEAP1 mutation status within lung adenocarcinoma. That is a statistical statement with no biological content, and R08 left its main caveat open: it could not rule out that the phenotype was really a smoking signature. This report addresses both, using all 392,489 retained probes rather than R08's 20,000 sample.

The phenotype is strong. A nested cross-validated classifier predicts KEAP1 mutation status from methylation alone at **AUROC 0.910** (95% CI 0.869–0.946) in 471 tumours with 84 mutants. The sex positive control through the identical pipeline reaches 1.000, and the result is stable across five seeds (0.868–0.910).

It is not a smoking signature. This was R08's declared-unresolvable limitation, and it turned out to be resolvable from the other direction. Defining the smoking signature in KEAP1-*wildtype* tumours only — which needs no never-smokers carrying the mutation — gives 37,008 probes, of which only **14,306 are shared** with the 115,557-probe KEAP1 signature. That is 12.4% of the KEAP1 signature and just **1.31**× what two independent signatures of these sizes would share by chance. Deleting every smoking-shared probe and retraining leaves the classifier at **AUROC 0.904**.

It has a coherent genomic structure. Differential probes are *depleted* in promoter CpG islands (21.5%, OR 0.54) and at transcription start sites (TSS200: 18.7%, OR 0.51), and *enriched* in open sea (34.6%, OR 1.44), island shelves (OR 1.26–1.27) and intergenic regions (OR 1.66). This is distal and gene-body remodelling, not promoter silencing.

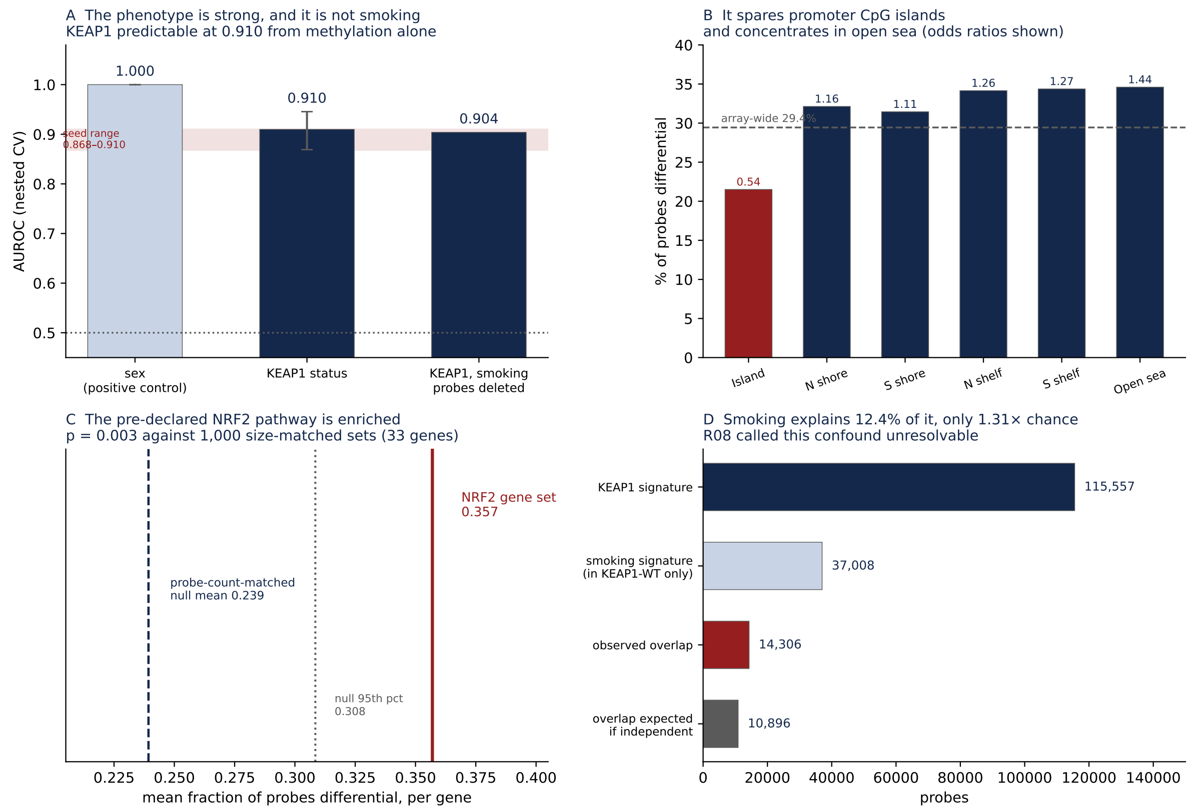
The pre-declared NRF2 pathway is enriched — and this was a genuine test, not a foregone conclusion. KEAP1 loss activates NRF2 post-translationally, which does not by itself predict any methylation change at NRF2 targets, so the prior was declared weak in advance. The 33 NRF2-pathway genes on the array average 35.7% of probes differential against a probe-count-matched null of 23.9% ($p = 0.003$).

1 Q1 — how strong is it?

	AUROC	95% CI
Sex (positive control, same pipeline)	1.000	1.000–1.000
KEAP1 mutation status	**0.910**	0.869–0.946
KEAP1, after deleting all 37,008 smoking-shared probes	0.904	—
Across 5 cross-validation seeds	median 0.898, range 0.868–0.910	

471 LUAD primary tumours, 84 KEAP1-mutant. LUSC was excluded entirely, because R08 established that pooling histologies confounds every comparison of this kind. Probe selection and hyperparameters were fitted strictly inside training folds.

The signature itself reproduces R08 almost exactly: 29.44% of all 392,489 probes against R08's 29.35% from a 20,000-probe sample, so R08's sampling error was negligible.



A Classifier performance, with the sex control and the smoking-deleted variant; shaded band is the 5-seed range. **B** Differential probes by CpG island context, odds ratios above each bar; red marks depletion. **C** The NRF2 gene set against 1,000 probe-count-matched random sets. **D** The overlap with smoking, against what chance alone would give.

2 Q2 — where in the genome?

Context	Probes	% differential	Odds ratio
CpG island	134,053	21.5%	**0.54**
North shore	52,248	32.1%	1.16
South shore	40,841	31.5%	1.11
North shelf	18,409	34.2%	1.26
South shelf	16,371	34.4%	1.27
Open sea	130,567	34.6%	**1.44**
by gene region			
TSS200	—	18.7%	**0.51**
TSS1500	—	27.2%	0.88
Gene body	—	31.0%	1.12
Intergenic (unannotated)	—	38.0%	**1.66**

Array-wide baseline is 29.4%; 390,543 of the 392,489 probes (99.5%) carry manifest annotation, so coverage is not a limitation here. The gradient is monotonic and unambiguous: the further from a promoter CpG island, the more likely a probe is differential. Whatever KEAP1 mutation associates with epigenetically, it is not classical promoter hypermethylation.

3 Q3 — which genes?

Of the 39 pre-declared NRF2/KEAP1-pathway genes, 33 have at least five probes on the array.

NRF2 pathway, mean fraction of probes differential		0.357
Unmatched random null (1,000 sets), mean	0.243	($p = 0.006$)
Probe-count-matched null (1,000 sets), mean	0.239	(**p = 0.003**)
Null 95th percentile		0.308

The audit re-ran the null matched on probes-per-gene, because random sets matched only on gene count would confound gene size with the result. The enrichment strengthened rather than weakened (p 0.006 → 0.003), and probes-per-gene correlates only weakly with the differential fraction in any case (Spearman $\rho = 0.089$).

Among genes with ≥10 probes, the most completely differential include *GATA5*, *LZTS1*, *UNCX*, *NCAN* and *GPX6* — developmental transcription factors and known methylation-regulated loci. This list is descriptive only: the ranking favours genes with dense probe coverage, and no gene-level claim is made from it.

4 Q4 — is it really smoking?

R08 stated this confound could not be resolved, because only 4 of 81 never-smokers carried a KEAP1 mutation, so no never-smoker stratum could support a genome-wide test. That framing assumed the test had to hold smoking constant. Reversing it removes the constraint: define the smoking signature among KEAP1-*wildtype* tumours (308 ever-smokers vs 65 never-smokers), then ask how much of the KEAP1 signature consists of those same probes.

KEAP1 signature	115,557 probes
Smoking signature (in KEAP1-wildtype tumours only)	37,008 probes
Observed overlap	14,306
Overlap expected if the two were independent	10,896
Observed / expected	**1.31**×
Share of the KEAP1 signature that is smoking-shared	**12.4%**
Jaccard index	0.104
Classifier AUROC after deleting all shared probes (355,481 of 392,489 probes retained)	**0.904** (from 0.910)

The two signatures are barely more similar than chance, and the classifier is indifferent to losing every probe they share. **The KEAP1 methylation phenotype is its own.**

The residual caveat, stated rather than hidden. The smoking comparison rests on 65 never-smokers, so the smoking signature is measured with less precision than its 37,008 probes suggests. If it is systematically under-detected, the true overlap is larger than 12.4%. Two things bound that concern: the limiting group sizes are comparable (65 never-smokers against 84 KEAP1 mutants), so the two signatures are not grossly unequally powered; and the deletion experiment in the audit does not depend on the smoking signature being complete — removing 37,008 candidate probes cost 0.006 AUROC.

5 What this changes

- **R08's positive finding is upgraded from statistical to characterised.** "29% of probes differ" becomes: AUROC 0.910, distal and intergenic rather than promoter-based, NRF2-enriched, and independent of smoking.

- **R08's limitation 3 is resolved, not merely acknowledged.** It was resolvable by inverting the design. That is worth noting for the other open confounds in this series.
- **The prognostic conclusion is unchanged.** R08 showed methylation adds nothing to mutation, stage and age for survival (LR $p = 0.85$), and nothing here revisits that. A strong biological phenotype and no prognostic value are not in conflict — KEAP1 mutation status is already known, so predicting it from a more expensive assay has no clinical use on its own.
- **Where it could matter.** The distal, NRF2-enriched pattern is a mechanistic lead about how KEAP1 loss reshapes the epigenome, and it is measurable in archival tissue. Whether it tracks NRF2 pathway *activity* — as opposed to KEAP1 genotype — is the natural next question, and needs matched expression data rather than more methylation.

6 Reproducing this report

- Analysis: `evidence/m11_keap1_phenotype.py`; audit: `evidence/audit_m11.py`
- Config hash, written before any outcome was read: `229f8cb6cc4312e8...1960db0f80`. The NRF2 gene set is listed inside it, fixed before any enrichment was computed
- Annotation: Illumina HumanMethylation450 v1.1 manifest, GEO GPL13534 supplementary file
- `evidence/keap1_signature_qvalues.csv.gz` — every one of the 115,557 significant probes with its q-value, so the signature can be re-used or checked directly
- Beta matrix: `gs://heydonto-quantara-lungcdx/data-request-2026-08/lung/ tcga_methylation_450k/` (see R08)
- Figure: `python3 make_figure.py`